

# Fahim Ahamed

Authorized to work in the U.S. and do not require sponsorship - U.S. Permanent Resident

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## Summary

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**Data Scientist** with a strong background in **AI Research**, delivering data-driven solutions across **healthcare**, **sustainability**, and **enterprise analytics**. Experienced in building **scalable ML pipelines**, conducting **advanced analytics**, and developing **predictive models** that drive measurable outcomes. Skilled in applying **NLP** and **computer vision** to production workflows. Proficient in **Python**, **PyTorch**, **TensorFlow**, **SQL**, and **MLOps**, with a proven ability to turn complex **data challenges** into actionable **business insights**.

## Work Experience

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### Research Assistant

Dec 2025 - Present

*CUNY Research Foundation*

*New York, NY*

- Developing an **agentic AI pipeline** using **RAG**, **LLMs**, and **MCP-connected tools** to enable natural language querying over complex geoscience databases.
- Bridging the gap between **geoscience domain expertise** and **AI engineering**, translating research requirements into technical specifications and iterating on pipeline outputs to ensure interpretability and scientific relevance.
- Designing the **end-to-end technical stack** spanning **vector database architecture**, domain-specific **embedding models**, **LLM optimization**, and **cloud deployment** to support a production-ready geoscience research tool.

### Data Evangelist Intern

Feb 2026 - May 2026

*DataCamp*

*New York, NY*

- Designed and maintained **scalable, reproducible pipelines** to collect, integrate, and validate **multi-source data** using **APIs** and **web scraping**.
- Built and published **data applications** and **analytical workflows**, transforming raw data into **user-facing tools** for exploring key metrics.
- Analyzed data to identify **trends, patterns**, and **key drivers**, producing clear outputs that support **data-driven decision-making**.

### Data Analyst

Oct 2025 - Feb 2026

*The City College of New York*

*New York, NY*

- Led analysis of **extensive institutional data** using **Excel**, **Tableau**, and **Python**, translating insights into actionable advising and engagement strategies.
- Conducted **data-driven surveys** and **retention studies**, uncovering behavioral patterns that inform targeted outreach and support programs.
- Built **analytical reports** and **visual dashboards** highlighting key trends, anomalies, and improvement areas, improving decision-making efficiency across departments.

## Technical Skills

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**Languages:** Python, SQL, R, Java, C/C++

**Data Analysis & Visualization:** NumPy, Pandas, SciPy, Matplotlib, Seaborn, Plotly, Tableau, Excel

**AI & Machine Learning:** TensorFlow, PyTorch, Scikit-learn, OpenCV, NLTK

**LLMs & Agents:** LLMs, RAG, MCP, Hugging Face, ChromaDB

**Database & Big Data:** MySQL, MongoDB, Snowflake

**MLOps & CI/CD Tools:** AWS, MLflow, Git/GitHub, Docker, Streamlit

**Others:** Data Preprocessing, Feature Engineering, Model Evaluation, A/B Testing

## Education

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**The City College of New York**

New York, NY

*M.S. in Data Science and Engineering*

*May 2027*

**East West University**

Dhaka, Bangladesh

*B.S. in Computer Science and Engineering*

*Dec 2023*

## Research Experience

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Explainable AI-Driven Hybrid Transfer Learning Framework for Accurate Skin Cancer Diagnosis – *Digital Health (SAGE)*

- Achieved **98.65% accuracy (F1 98.64%)** on multi-class skin cancer classification using a hybrid EfficientNetB0 + Random Forest model with advanced preprocessing, validated across datasets.

Robust Dual-Site Cancer Screening via Multi-Scale Vision Transformer and Rapid Recognition Pipeline – *IEEE Xplore*

- Developed a **multi-scale Vision Transformer framework** (Multi-ViT, MA-Transformer V2, MobileUNETR, TinyViT, Xception) achieving **99.12% accuracy** on PAD-UFES-20 and **99.37%** on Oral Cancer datasets, with strong F1 scores.

Texture Feature-Based Colonic Polyp Detection Using Deep Learning Techniques – *Springer Nature*

- **Achieved 97.33% IoU** for colonic polyp segmentation using DeepLabv3+, outperforming VGG16 and prior studies to set a new benchmark in colorectal cancer detection.

Optimizing Energy Usage: Genetic Algorithms Leading the Charge Toward Sustainability – *Springer Nature*

- **Reduced** annual household electricity usage by **34.54%** (90,449 watts) using **Genetic Algorithms**, demonstrating strong environmental impact and potential for **nationwide scalability** in sustainable energy optimization.

### Accepted Papers (In Press)

A Lightweight Transformer Ensemble for Explainable Depression Emotion and Severity Detection – *IEEE (2025)*

- Achieved **state-of-the-art F1 81.63%, Precision 83.24%, Specificity 86.92%** on benchmark datasets with a lightweight transformer ensemble, providing **LIME-based explanations** and a deployable real-time interface for depression detection.

### Papers Under Review

Multimodal Learning in Medical Imaging: Methods, Benchmarks, Evaluation, & Clinical Translation – *Elsevier*

- Conducted a structured review of **multimodal learning in medical imaging**, proposing a unified taxonomy while identifying key gaps in **robustness, reproducibility, and benchmark standardization** across healthcare AI systems.

Explainable Token-Fusion Transformer for Early Drowsiness State Recognition in Safety-Critical Systems – *Array Journal*

- Designed **EMTF-ViT**, achieving **99.2% intra-dataset accuracy** and **93%+ cross-dataset transfer**, with Grad-CAM interpretability and real-time low-power deployment for ADAS and safety applications.

## Data Projects

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**NYC HIV/AIDS Trend Analysis & Predictive Modeling** | *R, Tidyverse, ggplot2, sf, rpart, RandomForest, caret*

- Cleaned and engineered **multi-year NYC Open Data** on HIV/AIDS diagnoses, resolving suppressed values, harmonizing demographic fields, and producing analysis-ready datasets for modeling.
- Performed **spatiotemporal analysis of HIV/AIDS trends across NYC**, uncovering persistent geographic and demographic disparities, and leveraged **Random Forest models ( $R^2 = 0.65$ )** to identify high-impact predictors for targeted public health interventions.

**Global Population Growth Modeling & Forecasting** | *R, Tidyverse, ggplot2, leaps, caret*

- Engineered a **longitudinal time-series dataset** from World Bank indicators, integrating population, fertility, mortality, migration, and age structure data across **200+ countries**, and conducted **multi-decade demographic analysis** by region and income group.
- Developed and validated **population growth forecasting models**, achieving  **$R^2 = 0.755$** , and applied **rolling-window cross-validation** to assess short- and long-horizon predictive stability.

## AI/ML Projects

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**Clinical Risk Modeling & Unsupervised Structure Analysis** | *Python, NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn*

- Curated and standardized **clinical tabular data**, addressing missingness, outliers, and feature scaling to support supervised prediction and unsupervised structure discovery.
- Performed **diagnostic exploratory analysis** to identify risk-associated variables, then developed a **logistic regression classifier (80% test accuracy)** and evaluated **K-Means clustering with PCA and t-SNE projections** to assess latent structure in medical risk profiling.

**Intrusion Detection System** | *Python, NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn*

- **Processed and analyzed 2.5+ million network traffic records** through comprehensive EDA and reduced memory usage by **47.5%** through numerical datatype downcasting, improving training efficiency.
- **Achieved 97% median accuracy** in detecting network anomalies using machine learning models, uncovering key traffic patterns and strengthening anomaly detection capabilities.

## Additional

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- Solved **500+ problems** on platforms like **LeetCode, HackerRank, and Beecrowd** (formerly URI), combined.